



Hospital Engineering and Management

Lecture 2: Planning and Organization of Hospitals

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Learning Objectives

By the end of this lecture, students should be able to:

- ▶ explain the role of hospitals within the healthcare-delivery system;
- ▶ classify hospitals using common planning and service criteria;
- ▶ describe the major stages of hospital planning;
- ▶ identify important site-selection, zoning and circulation principles;
- ▶ explain the organizational structure of a hospital;
- ▶ relate biomedical-engineering requirements to early planning decisions.

Connection with the Previous Lecture

Lecture 1 Established

A hospital is a complex, safety-critical socio-technical system in which clinical, support, engineering and management services are interconnected.



Today: How should the hospital be planned and organized?

Role of a Hospital in Healthcare

A hospital is a coordinated institution that provides preventive, diagnostic, therapeutic, rehabilitative, educational and emergency services.

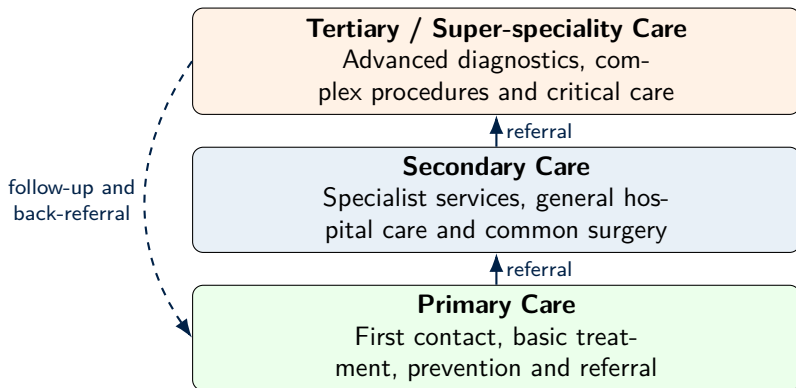
Direct Patient-Care Roles

- ▶ diagnosis and treatment;
- ▶ emergency and trauma care;
- ▶ inpatient and critical care;
- ▶ surgery and procedural care;
- ▶ rehabilitation and follow-up.

Wider Health-System Roles

- ▶ disease prevention and health promotion;
- ▶ teaching and workforce training;
- ▶ clinical research;
- ▶ public-health surveillance;
- ▶ disaster and outbreak response.

Hospitals within the Referral System



Planning Implication

The service package, equipment, staffing and bed capacity must match the hospital's position in the referral network.

Classification of Hospitals

Hospitals may be classified using more than one criterion.

By Ownership

- ▶ public or government;
- ▶ private for-profit;
- ▶ charitable or trust;
- ▶ teaching or institutional.

By Service Scope

- ▶ general hospital;
- ▶ speciality hospital;
- ▶ super-speciality hospital;
- ▶ teaching and research hospital.

By Length of Stay

- ▶ acute-care hospital;
- ▶ long-term or chronic-care facility;
- ▶ day-care or ambulatory facility.

By Size or Capacity

- ▶ small, medium or large;
- ▶ commonly expressed by sanctioned or operational beds.

What Is Hospital Planning?

Definition

Hospital planning is the systematic process of translating community health needs into a feasible combination of services, spaces, equipment, utilities, personnel, workflows and financial resources.

Need Assessment + Service Planning + Facility Design
+ Resource Planning = Functional Hospital

Key Principle

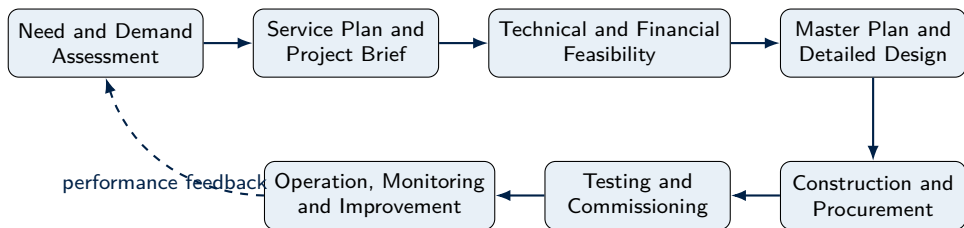
The building should follow the clinical and operational plan; the clinical plan should not be forced to fit an unsuitable building.

Major Inputs to Hospital Planning

- ▶ population and catchment area;
- ▶ disease burden and demographic profile;
- ▶ existing healthcare facilities;
- ▶ expected patient volume;
- ▶ proposed specialties and service levels;
- ▶ emergency and disaster responsibilities;
- ▶ site and environmental constraints;
- ▶ capital and operating budget;
- ▶ workforce availability;
- ▶ medical technology requirements;
- ▶ regulatory and safety requirements;
- ▶ future expansion and adaptability.

Planning begins with evidence, not with the drawing of rooms.

Hospital Planning Process



Engineering View

Each stage should define measurable requirements for reliability, safety, maintainability, utility capacity and equipment integration.

Need Assessment and Catchment Area

Questions to Answer

- ▶ Who will use the hospital?
- ▶ What illnesses and injuries are common?
- ▶ How many OPD, emergency and inpatient visits are expected?
- ▶ What services already exist nearby?
- ▶ Which unmet needs should the hospital address?

Typical Data Sources

- ▶ census and demographic data;
- ▶ epidemiological records;
- ▶ referral patterns;
- ▶ utilization statistics;
- ▶ stakeholder consultations;
- ▶ travel-time analysis.

Catchment area = the population and geographic region expected to use the hospital.

From Demand to Capacity

Capacity planning converts expected workload into beds, rooms, equipment and staffing.

Illustrative Bed Estimate

$$B \approx \frac{AL}{365 O}$$

B: beds, *A*: annual admissions, *L*: average length of stay,
O: target occupancy expressed as a fraction.

Also Estimate

- ▶ OPD consultation rooms;
- ▶ emergency treatment bays;
- ▶ operating rooms;
- ▶ ICU beds;
- ▶ imaging and laboratory workload;
- ▶ parking and waiting areas.

Caution

Capacity should include variability, peak demand, infection-isolation needs, maintenance downtime and future growth—not only average demand.

Site Selection for a Hospital

Accessibility and Community

- ▶ proximity to the intended population;
- ▶ road access and ambulance movement;
- ▶ public transport availability;
- ▶ travel time from referral centres;
- ▶ adequate parking and drop-off space.

Technical and Environmental Suitability

- ▶ reliable power, water and communication;
- ▶ drainage and waste-disposal options;
- ▶ soil, topography and flood risk;
- ▶ noise, air quality and nearby hazards;
- ▶ land for expansion and service access.

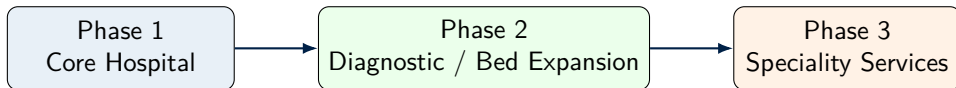
Planning Error to Avoid

A low-cost site can become expensive if utilities, access, drainage or expansion are inadequate.

Master Planning and Future Expansion

Master Plan

A master plan organizes the entire site—buildings, roads, utilities, open spaces, service yards, parking and future development—as one coordinated system.



- ▶ reserve expansion zones without blocking current operations;
- ▶ size utility corridors and plant rooms for staged growth;
- ▶ use modular layouts where appropriate;
- ▶ avoid placing permanent structures in future circulation paths.

Functional Zoning of a Hospital

Common Functional Zones

- ▶ public and outpatient zone;
- ▶ diagnostic and treatment zone;
- ▶ inpatient zone;
- ▶ critical-care and surgical zone;
- ▶ administrative zone;
- ▶ service and utility zone.

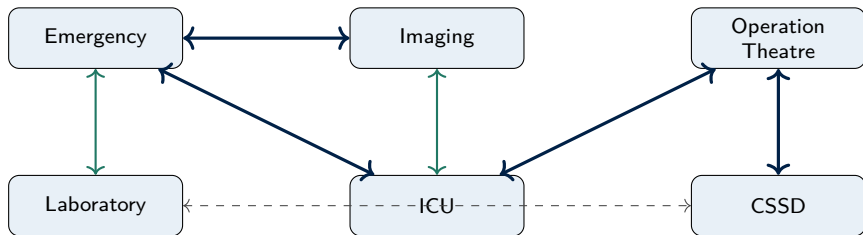
Why Zoning Matters

- ▶ improves wayfinding;
- ▶ protects privacy;
- ▶ reduces unnecessary travel;
- ▶ separates incompatible activities;
- ▶ supports infection control;
- ▶ simplifies security and access control.

Departments should be grouped by function, relationship and risk.

Departmental Relationships and Adjacency

Frequently interacting departments should be located close to each other.



Adjacency Analysis

Relationships may be rated as essential, desirable, acceptable or undesirable before the architectural layout is finalized.

Circulation and Flow Separation

Hospital planning must consider multiple simultaneous flows.

People Flows

- ▶ ambulatory patients;
- ▶ emergency and stretcher patients;
- ▶ inpatients;
- ▶ staff;
- ▶ visitors and attendants.

Material Flows

- ▶ sterile supplies and medicines;
- ▶ food and clean linen;
- ▶ specimens and blood products;
- ▶ soiled linen;
- ▶ biomedical waste and deceased persons.

Design Principle

Clean, sterile, public and contaminated flows should be separated wherever practical, with controlled crossings where separation is impossible.

Universal Access, Wayfinding and Patient Experience

- ▶ provide barrier-free access for wheelchairs, stretchers and mobility aids;
- ▶ use clear entrances, visible reception points and simple circulation routes;
- ▶ apply consistent signs, colours, symbols and floor numbering;
- ▶ minimize travel distance for frail and emergency patients;
- ▶ provide safe waiting areas, toilets, seating and drinking water;
- ▶ protect visual and acoustic privacy;
- ▶ include natural light, ventilation and calming environments where feasible.

Human-Centred Planning

A technically efficient hospital may still perform poorly if patients cannot find services, staff must walk excessive distances or privacy is compromised.

Biomedical Engineering Inputs during Planning

Biomedical engineers should participate before equipment is purchased or rooms are built.

- ▶ prepare equipment lists and technical specifications;
- ▶ estimate electrical load and outlet requirements;
- ▶ identify medical-gas and vacuum points;
- ▶ define space, structural and shielding needs;
- ▶ plan data connectivity and interoperability;
- ▶ verify environmental requirements;
- ▶ plan installation and service access;
- ▶ assess life-cycle cost and maintainability;
- ▶ coordinate acceptance testing and commissioning;
- ▶ plan replacement, training and spare support.

Example

CT and MRI rooms require early coordination of equipment access, cooling, power, shielding, structural loading and emergency provisions.

What Is Hospital Organization?

Definition

Hospital organization is the formal arrangement of authority, responsibility, communication and coordination through which hospital objectives are achieved.

Organization Must Define

- ▶ who makes decisions;
- ▶ who performs each function;
- ▶ who reports to whom;
- ▶ how departments coordinate;
- ▶ how performance is monitored.

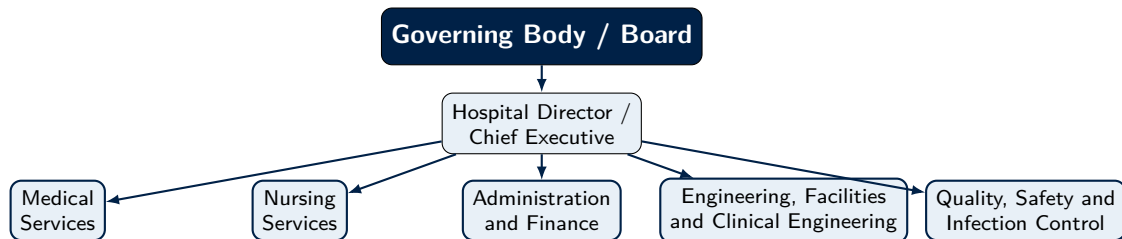
Without Clear Organization

- ▶ responsibilities overlap;
- ▶ critical tasks may be missed;
- ▶ communication becomes slow;
- ▶ accountability is weakened;
- ▶ patient safety may be affected.

Principles of Hospital Organization

- ▶ **Unity of objective:** all departments contribute to safe patient care;
- ▶ **Division of work:** duties are grouped by specialization and competence;
- ▶ **Authority with responsibility:** decision rights match assigned duties;
- ▶ **Unity of command:** each employee has a clearly defined reporting line;
- ▶ **Span of control:** one supervisor manages a practical number of staff;
- ▶ **Coordination:** clinical, support and engineering units work as one system;
- ▶ **Communication:** information moves accurately and rapidly;
- ▶ **Flexibility:** organization adapts to emergencies and changing technology;
- ▶ **Accountability:** performance and safety responsibilities are measurable.

Typical Hospital Organizational Structure



Important

The exact structure varies with hospital size, ownership, teaching status and service scope.

Line, Staff and Functional Relationships

Line Authority

Direct authority for achieving core objectives and supervising subordinates.

Example: Nursing superintendent to ward nursing staff.

Staff Authority

Advisory or specialist support without direct command over all departments.

Example: Legal, quality or infection-control advice.

Functional Authority

Limited authority over a specialized activity across departments.

Example: Biomedical engineering equipment-safety policy.

Good governance clarifies authority without creating departmental isolation.

Coordination through Hospital Committees

Committees bring multidisciplinary expertise to recurring safety and management decisions.

- ▶ infection prevention and control committee;
- ▶ quality and patient-safety committee;
- ▶ pharmacy and therapeutics committee;
- ▶ biomedical equipment committee;
- ▶ radiation-safety committee;
- ▶ fire and disaster-management committee;
- ▶ transfusion committee;
- ▶ ethics committee;
- ▶ occupational health and safety committee;
- ▶ waste-management committee.

Effective Committee Practice

Each committee needs a defined purpose, responsible members, meeting schedule, documented decisions, action owners and follow-up.

Planning Overview: Outpatient Department

Major Functional Elements

- ▶ reception and registration;
- ▶ waiting and patient information;
- ▶ consultation and examination rooms;
- ▶ minor procedure and treatment rooms;
- ▶ sample collection and pharmacy links;
- ▶ records, billing and support spaces.

Planning Priorities

- ▶ easy public access and clear wayfinding;
- ▶ separation of infectious patients where needed;
- ▶ privacy and acoustic control;
- ▶ balanced room and waiting capacity;
- ▶ efficient links to diagnostics;
- ▶ accessibility for wheelchairs and stretchers.

Planning Overview: Nursing Unit

A nursing unit is the basic organizational and spatial unit for inpatient care.

Typical Components

- ▶ patient rooms and isolation rooms;
- ▶ nurse station and documentation area;
- ▶ clean utility and medication area;
- ▶ dirty utility and waste holding;
- ▶ equipment and linen storage;
- ▶ staff, visitor and patient support spaces.

Design Goals

- ▶ direct observation and rapid response;
- ▶ short staff travel distances;
- ▶ safe medication and equipment handling;
- ▶ patient privacy and dignity;
- ▶ infection-control separation;
- ▶ reliable utilities and nurse-call systems.

Case Study: Planning a 200-Bed General Hospital

A growing city proposes a 200-bed general hospital with emergency, surgery, obstetrics, paediatrics, ICU, imaging and laboratory services.

Planning Questions

1. What community and demand data are required before fixing capacity?
2. Which departments need the strongest adjacencies?
3. Which patient and material flows should be separated?
4. What space and utility provisions are essential for future expansion?
5. At what stages should biomedical engineers participate?

Engineering Lesson

Bed count alone does not define a hospital. Service mix, workflow, utilities, staffing, equipment and expansion strategy must be planned together.

Classroom Activity: Develop a Functional Brief

Group Task

In groups, select one department: OPD, emergency, ICU, operation theatre, radiology or inpatient nursing unit.

Prepare a preliminary functional brief containing:

1. purpose and major activities;
2. users and expected workload;
3. essential rooms or spaces;
4. important departmental adjacencies;
5. equipment and utility requirements;
6. major hazards and safety provisions;
7. one provision for future expansion.

Summary

- ▶ Hospitals support treatment, prevention, training, research and emergency response.
- ▶ Hospital planning converts community needs into services, capacity, spaces and resources.
- ▶ Planning includes need assessment, feasibility, site selection, master planning, functional design, construction and commissioning.
- ▶ Zoning, adjacency and flow separation improve safety and operational efficiency.
- ▶ Hospital organization defines authority, responsibility, communication and coordination.
- ▶ Biomedical engineers should contribute from the earliest planning stage.

A well-planned hospital enables safe care before the first patient arrives.

Assignment for the Next Lecture

Select an existing hospital and prepare a one-page planning assessment.

1. identify its ownership, level of care, service scope and approximate capacity;
2. draw a simple functional-zoning or departmental-relationship diagram;
3. identify two good planning features;
4. identify two possible flow, access or expansion problems;
5. state three early-stage contributions expected from a biomedical engineer;
6. propose one practical planning improvement.

Submission Format

One page, individually prepared, including one labelled diagram and a short justification.

Suggested Reading

- ▶ G. D. Kunders, *Hospitals: Facilities Planning and Management*, Tata McGraw-Hill.
- ▶ B. M. Sakharkar, *Principles of Hospital Administration and Planning*, Jaypee Brothers Medical Publishers.
- ▶ S. Goel et al., *Textbook of Hospital Administration: A Problem-Solving Approach*, Elsevier India.

Next Lecture: Outpatient Services and Patient-Flow Planning

Thank You

Questions and Discussion